

Ewurakua Amoah 74492028,
Gabriel Kwesi Takyi Akurang 35882028,
Haris Tiymtaba Issah 41942028,
Emmanuel Kwesi Bentsil Odoom 35682028

//7.10 and 7.11

MariaDB [(none)]> create database hotelSystem;
Query OK, 1 row affected (0.009 sec)

MariaDB [(none)]> use hotelSystem;
Database changed

MariaDB [hotelSystem]> show tables;
Empty set (0.000 sec)

MariaDB [hotelSystem]> CREATE TABLE hotel (
-> hotelNo VARCHAR(5) NOT NULL,
-> hotelName VARCHAR(20) NOT NULL,
-> city VARCHAR(50),
-> PRIMARY KEY (hotelNo)
->);
Query OK, 0 rows affected (0.154 sec)

MariaDB [hotelSystem]> CREATE TABLE room (
-> roomNo INT NOT NULL CHECK (roomNo BETWEEN 1 AND 100),
-> hotelNo VARCHAR(5) NOT NULL,
-> type VARCHAR(20) NOT NULL CHECK (type IN ('Single', 'Double', 'Family')),
-> price DECIMAL(5,2) NOT NULL CHECK (price BETWEEN 10 AND 100),
-> PRIMARY KEY (roomNo, hotelNo),
-> FOREIGN KEY (hotelNo) REFERENCES hotel(hotelNo)
->);
Query OK, 0 rows affected (0.121 sec)

MariaDB [hotelSystem]> CREATE TABLE booking (
hotelNo VARCHAR(5) NOT NULL,
guestNo VARCHAR(5) NOT NULL, dateFrom DATE NOT NULL, dateTo DATE NOT
NULL, roomNo INT NOT NULL, PRIMARY KEY (hotelNo, guestNo, dateFrom),
FOREIGN KEY (hotelNo) REFERENCES hotel(hotelNo), FOREIGN KEY (guestNo)
REFERENCES guest(guestNo), FOREIGN KEY (roomNo, hotelNo) REFERENCES
room(roomNo, hotelNo));
Query OK, 0 rows affected (0.123 sec)

MariaDB [hotelSystem]> alter table Guest add Country varchar(50);
Query OK, 0 rows affected (0.267 sec)
Records: 0 Duplicates: 0 Warnings: 0

MariaDB [hotelSystem]> alter table Booking add PERIOD FOR booking_time(dateFrom,
dateTo);
Query OK, 0 rows affected (0.253 sec)
Records: 0 Duplicates: 0 Warnings: 0

MariaDB [hotelSystem]> alter table booking add CONSTRAINT unique_room_booking
UNIQUE (roomNo, hotelNo, booking_time WITHOUT OVERLAPS);
Query OK, 0 rows affected (0.256 sec)
Records: 0 Duplicates: 0 Warnings: 0

MariaDB [hotelSystem]> alter table booking add CONSTRAINT unique_guest_booking
UNIQUE (guestNo, booking_time WITHOUT OVERLAPS);
Query OK, 0 rows affected (0.243 sec)
Records: 0 Duplicates: 0 Warnings: 0

```

MariaDB [hotelSystem]> INSERT INTO hotel (hotelNo, hotelName, city) VALUES
-> ('H001', 'Grand Plaza', 'London'),
-> ('H002', 'Ocean View', 'Paris'),
-> ('H003', 'Sunset Stay', 'New York'),
-> ('H004', 'Eco Lodge', 'Tokyo'),
-> ('H005', 'Urban Ritz', 'Sydney'),
-> ('H006', 'Alpine Inn', 'Rio de Janeiro'),
-> ('H007', 'Metro Hub', 'Moscow'),
-> ('H008', 'Safari Tent', 'Mumbai'),
-> ('H009', 'Zen Resort', 'Beijing'),
-> ('H010', 'The Oasis', 'Cape Town'),
-> ('H011', 'Palace Stay', 'Berlin'),
-> ('H012', 'Breeze Hotel', 'Dubai'),
-> ('H013', 'Summit Lodge', 'Toronto'),
-> ('H014', 'Anchor Inn', 'Rome'),
-> ('H015', 'Golden Gate', 'Cairo');
Query OK, 15 rows affected (0.010 sec)
Records: 15  Duplicates: 0  Warnings: 0

```

```

MariaDB [hotelSystem]> INSERT INTO guest (guestNo, guestName, guestAddress,
Country) VALUES
-> ('G001', 'John Doe', '123 Main St', 'Brazil'),
-> ('G002', 'Jane Smith', '456 Oak Ave', 'Russia'),
-> ('G003', 'Carlos Silva', '789 Beach Rd', 'India'),
-> ('G004', 'Dmitry Ivanov', '10 Tverskaya', 'China'),
-> ('G005', 'Aarav Sharma', '45 MG Road', 'South Africa'),
-> ('G006', 'Li Wei', '88 Nanjing Rd', 'Brazil'),
-> ('G007', 'Thabo Molefe', '12 Mandela Way', 'Russia'),
-> ('G008', 'Alice Brown', '77 Baker St', 'India'),
-> ('G009', 'Bob Johnson', '55 Fifth Ave', 'China'),
-> ('G010', 'Emma Watson', '22 High St', 'South Africa'),
-> ('G011', 'Michael Chang', '33 Maple St', 'UK'),
-> ('G012', 'Sophia Rossi', '44 Via Roma', 'USA'),
-> ('G013', 'Amir Khan', '99 Sheikh Road', 'Germany'),
-> ('G014', 'Yuki Tanaka', '11 Shibuya St', 'France'),
-> ('G015', 'Elena Petrova', '66 Nevsky Blvd', 'Japan');
Query OK, 15 rows affected (0.001 sec)
Records: 15  Duplicates: 0  Warnings: 0

```

```

MariaDB [hotelSystem]> INSERT INTO room (roomNo, hotelNo, type, price) VALUES
-> (1, 'H001', 'Single', 45.00), (2, 'H001', 'Double', 75.00), (3, 'H001',
'Family', 95.00), (4, 'H001', 'Single', 49.00),
-> (1, 'H002', 'Single', 40.00), (2, 'H002', 'Double', 80.00), (3, 'H002',
'Family', 99.00), (4, 'H002', 'Double', 85.00),
-> (1, 'H003', 'Single', 55.00), (2, 'H003', 'Double', 88.00), (3, 'H003',
'Family', 100.00), (4, 'H003', 'Single', 58.00),
-> (1, 'H004', 'Single', 35.00), (2, 'H004', 'Double', 70.00), (3, 'H004',
'Family', 90.00), (4, 'H004', 'Double', 72.00),
-> (1, 'H005', 'Single', 60.00), (2, 'H005', 'Double', 95.00), (3, 'H005',
'Family', 100.00), (4, 'H005', 'Single', 62.00),
-> (1, 'H006', 'Single', 38.00), (2, 'H006', 'Double', 65.00), (3, 'H006',
'Family', 85.00), (4, 'H006', 'Family', 88.00),
-> (1, 'H007', 'Single', 42.00), (2, 'H007', 'Double', 78.00), (3, 'H007',
'Family', 95.00), (4, 'H007', 'Double', 82.00),
-> (1, 'H008', 'Single', 30.00), (2, 'H008', 'Double', 55.00), (3, 'H008',
'Family', 75.00), (4, 'H008', 'Single', 32.00),
-> (1, 'H009', 'Single', 50.00), (2, 'H009', 'Double', 85.00), (3, 'H009',
'Family', 100.00), (4, 'H009', 'Double', 89.00),
-> (1, 'H010', 'Single', 32.00), (2, 'H010', 'Double', 60.00), (3, 'H010',
'Family', 80.00), (4, 'H010', 'Single', 35.00),
-> (1, 'H011', 'Single', 45.00), (2, 'H011', 'Double', 75.00), (3, 'H011',
'Family', 95.00), (4, 'H011', 'Double', 78.00),

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```

-> (1, 'H012', 'Single', 58.00), (2, 'H012', 'Double', 90.00), (3, 'H012',
'Family', 100.00), (4, 'H012', 'Single', 60.00),
-> (1, 'H013', 'Single', 40.00), (2, 'H013', 'Double', 70.00), (3, 'H013',
'Family', 90.00), (4, 'H013', 'Double', 75.00),
-> (1, 'H014', 'Single', 48.00), (2, 'H014', 'Double', 82.00), (3, 'H014',
'Family', 98.00), (4, 'H014', 'Single', 50.00),
-> (1, 'H015', 'Single', 35.00), (2, 'H015', 'Double', 65.00), (3, 'H015',
'Family', 85.00), (4, 'H015', 'Family', 90.00);
Query OK, 60 rows affected (0.003 sec)
Records: 60 Duplicates: 0 Warnings: 0

```

```

MariaDB [hotelSystem]> INSERT INTO guest (guestNo, guestName, guestAddress,
Country) VALUES

```

```

-> ('G016', 'Lucas Mendes', '12 Paulista Ave', 'Brazil'),
-> ('G017', 'Olga Smirnova', '45 Arbat St', 'Russia'),
-> ('G018', 'Rohan Patel', '78 Link Rd', 'India'),
-> ('G019', 'Zhang Min', '15 Wangfujing', 'China'),
-> ('G020', 'Sipho Dlamini', '88 Zulu Way', 'South Africa'),
-> ('G021', 'Pedro Alvares', '34 Copacabana', 'Brazil'),
-> ('G022', 'Anna Kuznetsova', '90 Red Sq', 'Russia'),
-> ('G023', 'Priya Nair', '23 Marine Dr', 'India'),
-> ('G024', 'Wang Hao', '56 Bund Rd', 'China'),
-> ('G025', 'Zama Khumalo', '11 Durban Rd', 'South Africa'),
-> ('G026', 'Tom Hardy', '14 Piccadilly', 'UK'),
-> ('G027', 'Chris Evans', '402 Sunset Blvd', 'USA'),
-> ('G028', 'Leon Schmidt', '78 Kurfürstendamm', 'Germany'),
-> ('G029', 'Chloe Dubois', '92 Rue de Rivoli', 'France'),
-> ('G030', 'Kenji Sato', '5-1 Shinjuku', 'Japan');

```

```

Query OK, 15 rows affected (0.002 sec)
Records: 15 Duplicates: 0 Warnings: 0

```

```

MariaDB [hotelSystem]> INSERT INTO booking (hotelNo, guestNo, dateFrom, dateTo,
roomNo) VALUES

```

```

-> ('H001', 'G001', '2026-06-25', '2026-06-26', 1),
-> ('H001', 'G002', '2026-06-25', '2026-06-27', 2),
-> ('H002', 'G003', '2026-06-25', '2026-06-26', 1),
-> ('H002', 'G004', '2026-06-25', '2026-06-27', 2),
-> ('H003', 'G005', '2026-06-25', '2026-06-26', 1),
-> ('H003', 'G006', '2026-06-25', '2026-06-27', 2),
-> ('H004', 'G007', '2026-06-25', '2026-06-26', 1),
-> ('H004', 'G008', '2026-06-25', '2026-06-27', 2),
-> ('H005', 'G009', '2026-06-25', '2026-06-26', 1),
-> ('H005', 'G010', '2026-06-25', '2026-06-27', 2),
-> ('H006', 'G011', '2026-06-25', '2026-06-26', 1),
-> ('H006', 'G012', '2026-06-25', '2026-06-27', 2),
-> ('H007', 'G013', '2026-06-25', '2026-06-26', 1),
-> ('H007', 'G014', '2026-06-25', '2026-06-27', 2),
-> ('H008', 'G015', '2026-06-25', '2026-06-26', 1),
-> ('H008', 'G016', '2026-06-25', '2026-06-27', 2),
-> ('H009', 'G017', '2026-06-25', '2026-06-26', 1),
-> ('H009', 'G018', '2026-06-25', '2026-06-27', 2),
-> ('H010', 'G019', '2026-06-25', '2026-06-26', 1),
-> ('H010', 'G020', '2026-06-25', '2026-06-27', 2),
-> ('H011', 'G021', '2026-06-25', '2026-06-26', 1),
-> ('H011', 'G022', '2026-06-25', '2026-06-27', 2),
-> ('H012', 'G023', '2026-06-25', '2026-06-26', 1),
-> ('H012', 'G024', '2026-06-25', '2026-06-27', 2),
-> ('H013', 'G025', '2026-06-25', '2026-06-26', 1),
-> ('H013', 'G026', '2026-06-25', '2026-06-27', 2),
-> ('H014', 'G027', '2026-06-25', '2026-06-26', 1),
-> ('H014', 'G028', '2026-06-25', '2026-06-27', 2),
-> ('H015', 'G029', '2026-06-25', '2026-06-26', 1),
-> ('H015', 'G030', '2026-06-25', '2026-06-27', 2),

```

```

-> ('H001', 'G001', '2026-07-01', '2026-07-03', 3),
-> ('H002', 'G002', '2026-07-01', '2026-07-03', 3),
-> ('H003', 'G003', '2026-07-01', '2026-07-03', 3),
-> ('H004', 'G004', '2026-07-01', '2026-07-03', 3),
-> ('H005', 'G005', '2026-07-01', '2026-07-03', 3),
-> ('H006', 'G006', '2026-07-01', '2026-07-03', 3),
-> ('H007', 'G007', '2026-07-01', '2026-07-03', 3),
-> ('H008', 'G008', '2026-07-01', '2026-07-03', 3),
-> ('H009', 'G009', '2026-07-01', '2026-07-03', 3),
-> ('H010', 'G010', '2026-07-01', '2026-07-03', 3),
-> ('H011', 'G011', '2026-07-01', '2026-07-03', 3),
-> ('H012', 'G012', '2026-07-01', '2026-07-03', 3),
-> ('H013', 'G013', '2026-07-01', '2026-07-03', 3),
-> ('H014', 'G014', '2026-07-01', '2026-07-03', 3),
-> ('H015', 'G015', '2026-07-01', '2026-07-03', 3),
-> ('H001', 'G001', '2026-07-10', '2026-07-13', 4),
-> ('H002', 'G002', '2026-07-10', '2026-07-13', 4),
-> ('H003', 'G003', '2026-07-10', '2026-07-13', 4),
-> ('H004', 'G004', '2026-07-10', '2026-07-13', 4),
-> ('H005', 'G005', '2026-07-10', '2026-07-13', 4);

```

Query OK, 50 rows affected (0.002 sec)
Records: 50 Duplicates: 0 Warnings: 0

/ 7.11d

```

CREATE TRIGGER chk_booking_dates_update
BEFORE UPDATE ON Booking
FOR EACH ROW
BEGIN
    IF NEW.dateFrom <> OLD.dateFrom AND NEW.dateFrom <= CURDATE() THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Validation Error: New dateFrom must be in the
future.';
    END IF;

    IF NEW.dateTo <> OLD.dateTo AND NEW.dateTo <= CURDATE() THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Validation Error: New dateTo must be in the
future.';
    END IF;

    IF NEW.dateTo <= NEW.dateFrom THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Validation Error: dateTo must be greater than
dateFrom.';
    END IF;
END$$

```

DELIMITER ;
Query OK, 0 rows affected (0.192 sec)

```

MariaDB [hotelSystem]>
MariaDB [hotelSystem]> CREATE TRIGGER chk_booking_dates_update
-> BEFORE UPDATE ON Booking
-> FOR EACH ROW
-> BEGIN
->     IF NEW.dateFrom <> OLD.dateFrom AND NEW.dateFrom <= CURDATE() THEN
->         SIGNAL SQLSTATE '45000'
->         SET MESSAGE_TEXT = 'Validation Error: New dateFrom must be in the
future.';
->     END IF;
->
->     IF NEW.dateTo <> OLD.dateTo AND NEW.dateTo <= CURDATE() THEN
->         SIGNAL SQLSTATE '45000'

```

```

->          SET MESSAGE_TEXT = 'Validation Error: New dateTo must be in the
future.';
->      END IF;
->
->      IF NEW.dateTo <= NEW.dateFrom THEN
->          SIGNAL SQLSTATE '45000'
->          SET MESSAGE_TEXT = 'Validation Error: dateTo must be greater than
dateFrom.';
->      END IF;
->  END$$
Query OK, 0 rows affected (0.250 sec)

```

//7.12

```

MariaDB [hotelSystem]> create table archivebooking as select * from booking
where dateFrom <= '2026-06-25';
Query OK, 30 rows affected (0.138 sec)
Records: 30  Duplicates: 0  Warnings: 0

```

```

MariaDB [hotelSystem]> select * from archivebooking;

```

hotelNo	guestNo	dateFrom	dateTo	roomNo
H001	G001	2026-06-25	2026-06-26	1
H002	G003	2026-06-25	2026-06-26	1
H003	G005	2026-06-25	2026-06-26	1
H004	G007	2026-06-25	2026-06-26	1
H005	G009	2026-06-25	2026-06-26	1
H006	G011	2026-06-25	2026-06-26	1
H007	G013	2026-06-25	2026-06-26	1
H008	G015	2026-06-25	2026-06-26	1
H009	G017	2026-06-25	2026-06-26	1
H010	G019	2026-06-25	2026-06-26	1
H011	G021	2026-06-25	2026-06-26	1
H012	G023	2026-06-25	2026-06-26	1
H013	G025	2026-06-25	2026-06-26	1
H014	G027	2026-06-25	2026-06-26	1
H015	G029	2026-06-25	2026-06-26	1
H001	G002	2026-06-25	2026-06-27	2
H002	G004	2026-06-25	2026-06-27	2
H003	G006	2026-06-25	2026-06-27	2
H004	G008	2026-06-25	2026-06-27	2
H005	G010	2026-06-25	2026-06-27	2
H006	G012	2026-06-25	2026-06-27	2
H007	G014	2026-06-25	2026-06-27	2
H008	G016	2026-06-25	2026-06-27	2
H009	G018	2026-06-25	2026-06-27	2
H010	G020	2026-06-25	2026-06-27	2
H011	G022	2026-06-25	2026-06-27	2
H012	G024	2026-06-25	2026-06-27	2
H013	G026	2026-06-25	2026-06-27	2
H014	G028	2026-06-25	2026-06-27	2
H015	G030	2026-06-25	2026-06-27	2

30 rows in set (0.000 sec)

```

MariaDB [hotelSystem]> delete from booking where dateFrom <= '2026-06-25';
Query OK, 30 rows affected (0.002 sec)

```

```

MariaDB [hotelSystem]> select * from booking;

```

hotelNo	guestNo	dateFrom	dateTo	roomNo
H001	G001	2026-07-01	2026-07-03	3
H002	G002	2026-07-01	2026-07-03	3

H003	G003	2026-07-01	2026-07-03	3
H004	G004	2026-07-01	2026-07-03	3
H005	G005	2026-07-01	2026-07-03	3
H006	G006	2026-07-01	2026-07-03	3
H007	G007	2026-07-01	2026-07-03	3
H008	G008	2026-07-01	2026-07-03	3
H009	G009	2026-07-01	2026-07-03	3
H010	G010	2026-07-01	2026-07-03	3
H011	G011	2026-07-01	2026-07-03	3
H012	G012	2026-07-01	2026-07-03	3
H013	G013	2026-07-01	2026-07-03	3
H014	G014	2026-07-01	2026-07-03	3
H015	G015	2026-07-01	2026-07-03	3
H001	G001	2026-07-10	2026-07-13	4
H002	G002	2026-07-10	2026-07-13	4
H003	G003	2026-07-10	2026-07-13	4
H004	G004	2026-07-10	2026-07-13	4
H005	G005	2026-07-10	2026-07-13	4

20 rows in set (0.000 sec)

//7.13

```
MariaDB [hotelSystem]> CREATE VIEW cheapestHotel AS
-> SELECT
->     h.hotelName,
->     h.hotelNo,
->     h.City
-> FROM hotel h
-> JOIN room r ON h.hotelNo = r.hotelNo
-> GROUP BY
->     h.hotelNo,
->     h.hotelName,
->     h.City
-> ORDER BY AVG(r.price) ASC
-> LIMIT 10;
```

Query OK, 0 rows affected (0.091 sec)

//7.14

```
MariaDB [hotelSystem]> create view bricsGuests as select * from guest where
country in('Brazil' , 'Russia', 'India', 'China', 'South Africa');
```

Query OK, 0 rows affected (0.093 sec)

```
MariaDB [hotelSystem]> select * from bricsGuests;
```

guestNo	guestName	guestAddress	Country
G001	John Doe	123 Main St	Brazil
G002	Jane Smith	456 Oak Ave	Russia
G003	Carlos Silva	789 Beach Rd	India
G004	Dmitry Ivanov	10 Tverskaya	China
G005	Aarav Sharma	45 MG Road	South Africa
G006	Li Wei	88 Nanjing Rd	Brazil
G007	Thabo Molefe	12 Mandela Way	Russia
G008	Alice Brown	77 Baker St	India
G009	Bob Johnson	55 Fifth Ave	China
G010	Emma Watson	22 High St	South Africa
G016	Lucas Mendes	12 Paulista Ave	Brazil
G017	Olga Smirnova	45 Arbat St	Russia
G018	Rohan Patel	78 Link Rd	India
G019	Zhang Min	15 Wangfujing	China
G020	Sipho Dlamini	88 Zulu Way	South Africa
G021	Pedro Alvares	34 Copacabana	Brazil
G022	Anna Kuznetsova	90 Red Sq	Russia
G023	Priya Nair	23 Marine Dr	India

G024	Wang Hao	56 Bund Rd	China
G025	Zama Khumalo	11 Durban Rd	South Africa

20 rows in set (0.001 sec)

MariaDB [hotelSystem]> select * from cheapesthotel;

hotelName	hotelNo	City
Safari Tent	H008	Mumbai
The Oasis	H010	Cape Town
Grand Plaza	H001	London
Eco Lodge	H004	Tokyo
Golden Gate	H015	Cairo
Summit Lodge	H013	Toronto
Alpine Inn	H006	Rio de Janeiro
Anchor Inn	H014	Rome
Palace Stay	H011	Berlin
Metro Hub	H007	Moscow

//7.15

MariaDB [hotelSystem]> CREATE USER Manager;
Query OK, 0 rows affected (0.079 sec)

MariaDB [hotelSystem]> CREATE USER Director;
Query OK, 0 rows affected (0.022 sec)

MariaDB [hotelSystem]> CREATE USER Accounts;
Query OK, 0 rows affected (0.024 sec)

MariaDB [hotelSystem]> GRANT ALL PRIVILEGES
-> ON cheapestHotel
-> TO Manager, Director
-> WITH GRANT OPTION;
Query OK, 0 rows affected (0.031 sec)

MariaDB [hotelSystem]> GRANT ALL PRIVILEGES
-> ON bricsGuests
-> TO Manager, Director
-> WITH GRANT OPTION;
Query OK, 0 rows affected (0.027 sec)

// 7.16

MariaDB [hotelSystem]> GRANT SELECT
-> ON cheapestHotel
-> TO Accounts;
Query OK, 0 rows affected (0.028 sec)

MariaDB [hotelSystem]> GRANT SELECT
-> ON bricsGuests
-> TO Accounts;
Query OK, 0 rows affected (0.023 sec)

MariaDB [hotelSystem]> REVOKE SELECT
-> ON cheapestHotel
-> FROM Accounts;
Query OK, 0 rows affected (0.024 sec)

MariaDB [hotelSystem]> REVOKE SELECT

```

-> ON bricsGuests
-> FROM Accounts;
Query OK, 0 rows affected (0.028 sec)

```

//7.17

```

MariaDB [hotelSystem]> CREATE VIEW HotelBookingCount (hotelNo, bookingCount)
-> AS SELECT h.hotelNo, COUNT(*)
-> FROM Hotel h, Room r, Booking b
-> WHERE h.hotelNo = r.hotelNo AND r.roomNo = b.roomNo
-> GROUP BY h.hotelNo;
Query OK, 0 rows affected (0.101 sec)

```

```

MariaDB [hotelSystem]> SELECT *
-> FROM HotelBookingCount;

```

hotelNo	bookingCount
H001	20
H002	20
H003	20
H004	20
H005	20
H006	20
H007	20
H008	20
H009	20
H010	20
H011	20
H012	20
H013	20
H014	20
H015	20

15 rows in set (0.005 sec)

```

MariaDB [hotelSystem]> SELECT h.hotelNo, COUNT(*)
-> FROM Hotel h, Room r, Booking b
-> WHERE h.hotelNo = r.hotelNo AND r.roomNo = b.roomNo
-> GROUP BY h.hotelNo;

```

hotelNo	COUNT(*)
H001	20
H002	20
H003	20
H004	20
H005	20
H006	20
H007	20
H008	20
H009	20
H010	20
H011	20
H012	20
H013	20
H014	20
H015	20

15 rows in set (0.004 sec)


```

MariaDB [hotelSystem]> SELECT hotelNo
-> FROM HotelBookingCount
-> WHERE hotelNo ='H001';

```

```

+-----+
| hotelNo |
+-----+
| H001    |
+-----+
1 row in set (0.004 sec)

```

```

MariaDB [hotelSystem]> SELECT h.hotelNo
-> FROM Hotel h, Room r, Booking b
-> WHERE h.hotelNo = r.hotelNo AND r.roomNo = b.roomNo
-> GROUP BY h.hotelNo
-> HAVING h.hotelNo = 'H001';

```

```

+-----+
| hotelNo |
+-----+
| H001    |
+-----+
1 row in set (0.001 sec)

```

```

MariaDB [hotelSystem]> SELECT MIN(bookingCount)
-> FROM HotelBookingCount;

```

```

+-----+
| MIN(bookingCount) |
+-----+
|                20 |
+-----+
1 row in set (0.007 sec)

```

```

MariaDB [hotelSystem]> SELECT COUNT(*)
-> FROM Hotel h, Room r, Booking b
-> WHERE h.hotelNo = r.hotelNo AND r.roomNo = b.roomNo
-> GROUP BY h.hotelNo
-> ORDER BY COUNT(*) ASC
-> LIMIT 1;

```

```

+-----+
| COUNT(*) |
+-----+
|        20 |
+-----+
1 row in set (0.002 sec)

```

```

MariaDB [hotelSystem]> SELECT COUNT(*)
-> FROM HotelBookingCount;

```

```

+-----+
| COUNT(*) |
+-----+
|        15 |
+-----+
1 row in set (0.004 sec)

```

```

MariaDB [hotelSystem]> SELECT COUNT(DISTINCT h.hotelNo) as Count
-> FROM Hotel h, Room r, Booking b
-> WHERE h.hotelNo = r.hotelNo AND r.roomNo = b.roomNo;

```

```

+-----+
| Count |
+-----+
|      15 |
+-----+
1 row in set (0.003 sec)

```

```
MariaDB [hotelSystem]> SELECT hotelNo
-> FROM HotelBookingCount
-> WHERE bookingCount > 1000;
Empty set (0.004 sec)
```

```
MariaDB [hotelSystem]> SELECT h.hotelNo
-> FROM Hotel h, Room r, Booking b
-> WHERE h.hotelNo = r.hotelNo AND r.roomNo = b.roomNo
-> GROUP BY h.hotelNo
-> HAVING COUNT(*) > 1000;
Empty set (0.002 sec)
```

```
MariaDB [hotelSystem]> SELECT hotelNo
-> FROM HotelBookingCount
-> ORDER BY bookingCount;
```

```
+-----+
| hotelNo |
+-----+
| H001    |
| H009    |
| H002    |
| H010    |
| H003    |
| H011    |
| H004    |
| H012    |
| H005    |
| H013    |
| H006    |
| H014    |
| H007    |
| H015    |
| H008    |
+-----+
```

15 rows in set (0.001 sec)

```
MariaDB [hotelSystem]> SELECT h.hotelNo
-> FROM Hotel h, Room r, Booking b
-> WHERE h.hotelNo = r.hotelNo AND r.roomNo = b.roomNo
-> GROUP BY h.hotelNo
-> ORDER BY COUNT(*) ASC;
```

```
+-----+
| hotelNo |
+-----+
| H001    |
| H009    |
| H002    |
| H010    |
| H003    |
| H011    |
| H004    |
| H012    |
| H005    |
| H013    |
| H006    |
| H014    |
| H007    |
| H015    |
| H008    |
+-----+
```

15 rows in set (0.003 sec)

// 7.19

The materialized view can be maintained without accessing Supplier or Part when a new supplier row is inserted for a partNo already in the view, or when a deleted row's partNo still exists in the other table. Both base tables only need to be accessed when a partNo is updated, requiring verification that the join condition still holds.